

3. DATA GATHERING / DATA USED

3.1 Overview of Platform Datasets & Entities

Seva Sankalp is an interactive peer-to-peer resource redistribution web platform. Rather than analyzing static third-party datasets, the application manages multi-tenant relational data models generated by registered community donors, grassroots recipients, and verified charitable NGOs. The core data entities captured, preprocessed, and maintained within the SQLite/PostgreSQL relational database include:

- **User Profile Data:** Registered accounts, authentication secrets, role permissions, and e-KYC status.
- **Resource Data:** Physical goods catalog, descriptions, conditions, and spherical GPS coordinates.
- **Request Data:** Beneficiary donation requests, approval lifecycles, and anti-flipping cooldown logs.
- **Statutory NGO Data:** Government NITI Aayog Darpan IDs, Organization PAN numbers, and verified UPI VPAs.
- **Donation History Data:** Audit logs of fulfilled exchanges, mutual ratings, and photographic impact proofs.
- **Points Ledger Data:** Append-only transactions recording earned and redeemed Karma points.
- **Handover Communication Data:** In-app secure chat logs coordinating physical item handovers.

Sample Users Table:

id	email	role	verification.stat us	trust.sc ore	points. balance	created.at
1	admin@sevasankalp.org	admin	approved	100	0	2026-06-08 09:42:09
2	donor@gmail.com	user	approved	48	350	2026-06-08 09:42:09
3	receiver@gmail.com	user	approved	25	100	2026-06-08 09:42:10
4	anish.sandaka1806@gmail.com	user	approved	85	450	2026-06-08 09:56:33
5	srinivas.sandaka75@gmail.com	user	approved	60	200	2026-07-05 11:12:10

Sample Resources Table:

id	donor.id	title	category	condition	location.lat	location.lng	status	created.at
1	4	rice (1kg)	Food	Good	17.9107	83.4538	Available	2026-07-02 14:58:25
2	5	NCERT class 10 science	Books	Fair	18.0184	83.5709	Available	2026-07-05 12:19:05
3	4	wheat flour (1kg)	Food	Good	18.2773	83.8950	Available	2026-07-06 09:50:54
4	2	Lenovo ThinkPad Laptop	Electronics	Good	18.1067	83.3978	Fulfilled	2026-08-10 10:15:30

Sample Resource Requests Table:

id	resource.id	receiver.id	status	created.at	updated.at
1	1	3	Fulfilled	2026-07-03 14:30:15	2026-07-05 11:25:37
2	2	3	Fulfilled	2026-07-05 12:45:00	2026-07-06 14:20:10
3	4	5	Fulfilled	2026-08-11 09:15:20	2026-08-14 16:30:45
4	3	5	Pending	2026-08-20 11:00:00	2026-08-20 11:00:00

Sample Statutory NGO Profiles Table:

id	ngo.name	darpan.id	org.pan	upi.id	verification.status	created.at
6	Global Heart Foundation	AP/2021/0284910	AABTG4921E	globalheart@upi	approved	2026-06-28 05:25:14
7	Team Water Welfare Trust	AP/2022/0319482	AAATE1234F	9848294904@ptyes	approved	2026-06-29 14:57:00
10	Vidya Jyothi Rural Trust	DL/2020/0192841	AAATV9812K	vidyajyothi@upi	approved	2026-08-01 11:15:00

Sample Donation History Table:

id	request.id	donor.rating	receiver.rating	receipt.photo	impact.message	completed.at
1	1	5.0	5.0	uploads/impact/receipt_1.jpg	Ration received, helped family during exam week.	2026-07-05 11:25:37

id	request.id	donor.rating	receiver.rating	receipt.photo	impact.message	completed.at
2	2	5.0	4.8	uploads/impact/receipt_2.jpg	Received class 10 science textbooks. Thank you!	2026-07-06 14:20:10
3	3	5.0	5.0	uploads/impact/receipt_3.jpg	Working laptop received for engineering studies.	2026-08-14 16:30:45

Sample Points Transaction (Karma Ledger) Table:

id	user.id	amount	transaction.type	description	created.at
1	4	50	earned_donation	Fulfilled donation of 1kg rice	2026-07-05 11:25:37
2	5	50	earned_donation	Fulfilled donation of NCERT textbooks	2026-07-06 14:20:10
3	2	50	earned_donation	Fulfilled donation of ThinkPad laptop	2026-08-14 16:30:45

Sample Handover Messages Table:

id	request.id	sender.id	content	created.at
1	1	3	Hello, is this item available for pickup near Pusapatirega?	2026-07-03 14:31:00
2	1	4	Yes, available! Can we meet tomorrow evening at 5 PM?	2026-07-03 14:35:20
3	1	3	Perfect, 5 PM at RTC Complex junction works for me.	2026-07-03 14:36:10

3.2 Preprocessing and Data Cleaning

- **Coordinate Clamping:** GPS latitude and longitude pairs submitted through Leaflet map clicks are parsed as 64-bit floating point numbers and validated within real geographical bounds (-90 to +90 and -180 to +180).
- **Image Resizing & EXIF Stripping:** Uploaded listing images and impact receipts are downsampled via Pillow (PIL) to maximum 800x800 resolution at 80% JPEG compression, automatically removing embedded GPS EXIF metadata to protect donor home privacy.
- **Format Regularization:** NITI Aayog NGO Darpan IDs (e.g. DL/2021/0123456) and Organization PAN numbers (10 alphanumeric characters) are converted to uppercase and checked against strict regex patterns prior to persistence.